KAYITARE STEVEN

ARTIFICIAL INTELLIGENCE UNDERGRADUATE | COMPUTER VISION & DEEP LEARNING

Gangwon, South Korea | +82 10-9631-2302 | stevenkayitare707@gmail.com
LinkedIn: linkedin.com/in/kayitare-steven | GitHub: github.com/stevenkayitaresteven | Portfolio: github.com/stevenkayitaresteven/portifolio

RESEARCH PROFILE

Artificial Intelligence undergraduate at Kyungdong University with a strong academic foundation in calculus, probability and statistics, algebra, and neural computation. Builds applied AI systems across computer vision, NLP, and multimodal learning, with hands-on experience in PyTorch, Hugging Face, model fine-tuning, evaluation, and deployment. First author of an LLM-evaluation research manuscript and contributor to the PyTorch open-source codebase. Interested in research-driven development of visual representation learning and multimodal intelligent systems.

RESEARCH INTERESTS

Computer vision and visual representation learning • Multimodal perception • Deep learning • Generative models • 3D data and visual understanding

EDUCATION & ACADEMIC PERFORMANCE

Kyungdong University — Global Campus, South Korea

B.S. in Artificial Intelligence | *Aug 2024 – Expected 2028*

GPA: 3.86/4.50 | Academic percentile: 92.6/100 | Rank: 8th of 295 students

Selected coursework: Calculus (A+), Probability & Statistics (A+), Algebra (A+), Neurocomputing (A+), Machine Learning (A0), Computer Vision (A0), Numerical Analysis (A0), Natural Language Processing (B+), Image & Video Analysis (B+), Neural Networks (B0).

TECHNICAL SKILLS

Programming: Python, C++, SQL, JavaScript/TypeScript, Bash
Machine Learning: PyTorch, TensorFlow, scikit-learn, model training, transfer learning, evaluation
Computer Vision: OpenCV, image/video analysis, ResNet-50, visual classification, OCR, ONNX
NLP & Multimodal AI: Hugging Face Transformers, LLMs, Whisper ASR, text-image-audio/video pipelines
Fine-tuning & Evaluation: LoRA, QLoRA, PEFT, TRL, prompt engineering, LLM-as-judge, precision/recall/F1
Engineering & Tools: FastAPI, Docker, Git/GitHub, Linux, data pipelines, MinHash/LSH deduplication, AWS
Mathematical Foundations: Linear algebra, calculus, probability and statistics, numerical analysis, optimization

SELECTED RESEARCH & ENGINEERING PROJECTS

Sentinel — Multimodal Content-Moderation Engine

Python • PyTorch • Hugging Face • OpenCV • Whisper • FastAPI • Docker

- Developed an end-to-end pipeline for moderating text, images, video, audio, and files across a 12-category safety taxonomy, producing per-input actions and confidence scores.
- Combined transformer-based NLP detectors with offline rule-based safeguards, including lexical checks, PII pattern detection, and file-type inspection, to support graceful degradation.
- Integrated image detection, Whisper-based audio transcription, video-frame sampling, and OCR into a shared result schema behind a unified moderation API.
- Built data-preparation workflows for PII scrubbing, MinHash/LSH deduplication, and weak labeling; prepared QLoRA/PEFT fine-tuning recipes across 12 public datasets. Packaged the service with Docker and a 140+ test suite.

  Live demo: huggingface.co/spaces/Kstevo34/sentinel-chat

The Longest-Answer Heuristic — LLM Evaluation Research

First Author • Python • LaTeX • Experimental Design • Statistical Analysis

- Authored a research manuscript investigating answer-length bias in LLM-generated multiple-choice questions and its implications for assessment integrity.
- Designed controlled experiments to isolate answer length as a potential confounding signal; manuscript is in submission following adversarial peer review and a publication-readiness audit.

  Research repository: github.com/stevenkayitaresteven/The-Longest-Answer-Heuristic

Crop-Pathology Classifier — ResNet-50 Transfer Learning

PyTorch • Computer Vision • Transfer Learning • PlantVillage

- Fine-tuned a ResNet-50 model to classify crop diseases across 38 PlantVillage classes and produced a training pipeline, technical report, and conference poster.

KigaliElite Realty / Imbata — AI-Enabled Real-Estate Platform

Node.js • FastAPI • SQLAlchemy • MongoDB • LLM-based Extraction

- Built a real-estate aggregation platform with an automated listing workflow combining scheduled scraping and LLM-assisted information extraction, alongside a responsive interface for the Kigali market.

RESEARCH & OPEN-SOURCE CONTRIBUTIONS

- PyTorch — Authored a bug-fix pull request in the torch/accelerator module, following project contribution conventions for source changes, commit messaging, and PR description.

ADDITIONAL EDUCATION

GS Remera Rukoma High School, Rwanda | Diploma in Mathematics, Physics & Computer Science (MPC), 2019–2023

CERTIFICATIONS

Microsoft Applied Skills — Azure AI Foundry (MCP + Agents) • AI Agents Fundamentals • Claude 101 & Code 101

LANGUAGES

English (Full Professional) • Kinyarwanda (Native) • Swahili (Professional) • Korean (Limited Working)